

The transshipment problem

Total unimodularity of the incidence matrix

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Optimization: principles and algorithms

Total unimodularity of the incidence matrix

Total unimodularity of the matrix defining the constraints of a linear optimization problem guarantees that the basic solutions are integer if the data are integer.

And, therefore, the optimal solution is also integer

This is a great property, given the difficulty of discrete optimization.

Transshipment problem in standard form

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$

subject to

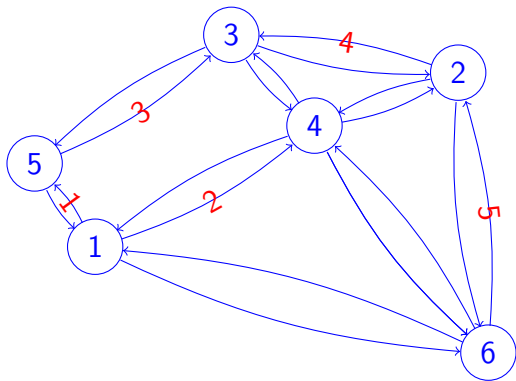
$$\sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = s_i, \forall i \in \mathcal{N},$$
$$x_{ij} \geq 0, \forall (i,j) \in \mathcal{A}.$$

subject to

$$\min_{x \in \mathbb{R}^n} c^T x$$

$$Ax = s,$$
$$x \geq 0.$$

Incidence matrix



	1	2	3	4	5	...
1	1	1	0	0	0	
2	0	0	0	1	-1	
3	0	0	-1	-1	0	
4	0	-1	0	0	0	
5	-1	0	1	0	0	
6	0	0	0	0	1	

Theorem 22.6

The incidence matrix of a network is totally unimodular.

Proof

- By contradiction, assume that it is not the case.

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- ▶ Number of non-zero entries in a column of B : 0, 1 or 2.

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- ▶ There exist square matrices with determinant $\notin \{0, 1, -1\}$.
- ▶ Among all of them, consider the one with minimum size k : B .
- ▶ Note that $k > 1$.
- ▶ Number of non-zero entries in a column of B : 0, 1 or 2.
- ▶ We treat each case separately.

No non-zero entry

Context

- ▶ B is $k \times k$.
- ▶ $\det(B) \notin \{0, 1, -1\}$.
- ▶ $\forall \ell \times \ell$ submatrices B' : $\det(B') \in \{0, 1, -1\}$.
- ▶ One column of B contains only zeros.

*Impossible, because it would mean that $\det(B) = 0$.
Therefore, no column of B contains only 0s.*

One non-zero entry

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- ▶ $\det(B) \notin \{0, 1, -1\}$.
- ▶ $\forall \ell \times \ell$ submatrices B' : $\det(B') \in \{0, 1, -1\}$.
- ▶ One column of B contains only one non-zero entry.

$$B = \begin{pmatrix} b_{11} & b_{12} \cdots b_{1k} \\ 0 & \\ \vdots & B' \\ 0 & \end{pmatrix}$$

$$\det(B) = b_{11} \det(B') = \det(B') \text{ or } -\det(B')$$

Impossible as B is the smallest matrix.

Two non-zero entries

Context

- ▶ B is $k \times k$.
- ▶ $\det(B) \notin \{0, 1, -1\}$.
- ▶ $\forall \ell \times \ell$ submatrices B' : $\det(B') \in \{0, 1, -1\}$.
- ▶ Each column of B contains exactly two non-zero entries.

The two non zero entries are 1 and -1.

If you sum the elements of each column, it is equal to zero.

So, summing all the rows of B gives 0

B is singular, and its determinant is zero.

Contradiction.

Corollary

If the supply vector s of the transshipment problem is integer, that is, if $s \in \mathbb{Z}^m$, and if the problem is bounded, then it has an optimal solution which is integer.

Summary

The incidence matrix of a network is totally unimodular

As the constraint matrix of the transshipment problem is the incidence matrix of the network, we have the guarantee to obtain integer optimal solutions if the demand and supply data are integer.

If you send trucks, there is no need to break them into pieces, or to rely on discrete optimization.